

1 Introduction

Decision problems in urban transportation systems can have huge socio-economic impact, which need careful analysis in various levels of resolution and an efficient optimization framework. In general, these systems are complex (e.g., dynamic, stochastic, non-linear) and difficult to model analytically, which precludes the use of traditional optimization methods. A powerful alternative is simulation-enabled approach. Recently, a variety of commercial or open-source traffic simulation tools have been developed, providing us with a more flexible and comprehensive way to capture the intricate interactions between the supply and the demand components of the mobility networks. These tools are therefore often used to evaluate and compare a set of predetermined solutions (e.g., traffic facility locations or network design plans) in the context of what-if analysis or sensitivity analysis for better decision-making. However, when the number of feasible or candidate solutions becomes large, it is impractical or prohibitive to simulate all of them. This is common in urban transportation problems which involve computationally expensive calibrated traffic simulation model and high dimensional decision variables. For example, the network signal timing problem needs to optimize the green times of all the phases at all the intersections in the network (Osorio and Bierlaire, 2013, Zheng et al. 2019, 2021), leading to a huge number of feasible solutions, which generally grows exponentially in problem dimension (Li et al., 2015). Simple enumeration is even less applicable when simulations are stochastic, as each simulation run only gives a noisy observation of the underlying quality of a solution. For such problems with large number of alternatives or measurement noises, research framework of simulation-based optimization (SO) was proposed, which design algorithms that seek good solutions quickly. SO has been an increasingly hot research topic, see relevant reviews (***).

In addition to the abovementioned challenges, urban transportation problems often involve multiple performance measures or criteria (Zheng 2019), such as traffic efficiency, safety, environmental impact and equity. Some of these performance measures are the primary objectives to be optimized, such as the total travel time on the network, whereas others have to satisfy some control targets such as the total traffic emissions on the network. Moreover, these performance measures are typically stochastic with very limited prior information about their distributions, making the problem harder. So, there is an urgent need for novel algorithms tackling stochastic constrained multi-objective problems arise in under transportation systems. To this end, this study will focus on urban transportation problems in the following form:

$$\begin{aligned} \min_{\mathbf{x} \in \mathcal{X}} & \left[\mathbb{E}_{\xi} [f_1(\mathbf{x}, \xi)], \mathbb{E}_{\xi} [f_2(\mathbf{x}, \xi)] \right] \\ \text{s. t. } & \Pr_{\xi} [f_3(\mathbf{x}, \xi) - \tau \leq 0] \geq 1 - \epsilon \end{aligned} \quad (1)$$

Functions $f_1(\mathbf{x}, \xi)$ to $f_3(\mathbf{x}, \xi)$ are three scalar-valued stochastic performance measures output from traffic simulator, which depend on the decision vector $\mathbf{x}$ and the random parameter vector ξ in the traffic simulator. Suppose $\mathbf{x}$ lies in a known feasible set $\mathcal{X}$, and ξ has a known distribution $p(\xi)$ with bounded support Ξ . Depending on what underlying system stochasticity needs to be captured for the problem, ξ can contain uncertain parameters in traffic demand, driving behavior, routing behavior, etc. $\mathbb{E}_{\xi}$ and $\Pr_{\xi}$ denote the expectation and the probability with respect to ξ , respectively. The constraint means the probability that $f_3(\mathbf{x}, \xi)$ is over the limit τ is at most $\epsilon \in (0,1)$, where τ and ϵ are

predetermined scalars. Problem (1) is therefore to find the nondominated solutions in terms of their expected f_1 and f_2 values with their f_3 values satisfy the probabilistic constraint.

Generally, there are three ways to address the SO problems: direct-search methods, stochastic gradient methods and metamodel methods (Osorio and Bierlaire 2013). Note functions f_1 to f_3 in problem (1) are all stochastic and costly to evaluate and may require different procedures or components in simulations. Thus, an ideal SO algorithm should make full use of the information given by the simulation observations, i.e., infer useful information globally with a local observation. In this regard, direct-search and stochastic gradient methods are not the best choice, since accurate estimation of search directions or gradients is sample-intensive and only provide local information. Thus, in our algorithm design, we focus on metamodels to approximate the expectations in the objectives and the probability in the constraint of problem (1). The metamodels are iteratively fitted using the simulated observations and provide improved prediction of the underlying performance measures globally to guide the search of the feasible Pareto optimal solutions. In fact, metamodel-guided sampling policies have been proposed extensively in SO literature. For instance, kriging model is used with expected improvement (EI) sampling (Forrester and Keane 2009), RBF model with scoring criterion search (Müller et al. 2014), polynomials with trust-region search (Osorio and Bierlaire 2013), Gaussian process regression (GPR) with Bayesian optimization (BO) (e.g., Scott et al. 2011), etc. Naturally, the applicability and the performance of the algorithm rely on what combination of metamodel and sampling policy is used. We propose parametric-GPR-based approximate-KG sampling policy for tackling challenging problem (1), which is the main contribution of this study.

BO is a powerful SO framework which starts with a prior belief (estimates of the function values and the uncertainties) on the solutions and selects the new sample which maximizes the value of the measurement, the new observation is then used for updating the posterior belief. The value of measurement is called the acquisition function defined by the sampling policy. In particular, the KG policy defines acquisition function as the expected improvement in the *best posterior mean* (known as the KG-factor). KG policy has theoretical convergence guarantee and perform well in practice (e.g., Negoescu et al. 2011, Tan and Gao 2016, Wu and Frazier 2016). GPR is commonly used in BO due to neat posterior update of the Gaussian process. Many popular metamodels have similar essence as GPR, such as various kriging models that estimate spatial correlations (**), and the Gaussian Markov random fields which restrict GPR to have certain dependence structure among the solutions (Semelhago, et al. 2019). Advantages of the GPR-based KG policy include: 1) convenient KG-factor computation with explicit modeling of possible correlations among solutions (Frazier 2009); 2) flexibility in parameterizing the prior as linear combinations of any basis functions (or features) of the solution (Powell and Ryzhov 2009); and 3) available techniques for accelerating the computation of the sampling decision when solution space is large (Ryzhov and Powell 2009, Tan and Gao 2016). We design parametric GPR models for f_1 to f_3 with problem-specific feature vector of the solution, which makes computation affordable by maintaining posteriors only on model coefficients and the sampled solutions rather than on all the solutions. In addition, we extend the GPR based KG policy to multiobjective setting by combining the posterior distribution and the deterministic multiobjective solvers in generating candidate samples

for the next iteration.

Our second contribution is designing a novel updating scheme for learning unknown and heterogeneous variances of the solutions. In urban transportation problems, the variance of the same performance measure can vary notably for different solutions, as observed, for example, in Osorio and Bierlaire (2013) and Tan and Gao (2016). This necessitates effective learning of the variances especially considering the fact that most solutions will not be sampled under limited budget. In our approach, we divide the feature space of the solutions into grids, then we estimate the variance of any unsampled solution $\mathbf{x}$ by the average of those of the sampled solutions whose features lies in the same grid as $\mathbf{x}$. This grid-based learning approach is similar to that proposed by Salemi et al. (2019), but they defined grids directly on the solution space, which can be costly to manage when problem dimension grows. Our variance estimation approach enables cheap approximation of the KG-factor, which is one research focus of the KG policy (Qu 2015, Zhang 2016) since exact Bayesian inferences is impossible with single observation with a multivariate Gaussian prior. Also note that our approach for learning under measurement noises is quite different with those proposed for cases with relatively small number of alternatives, e.g., sample average approximation (SAA) (***), simulation allocation rule such as OCBA (Li et al., 2015), shrinking ball method (Fan and Hu, 2018), etc., where each alternative is sampled for multiple times and variances are typically known.

Finally, this study enriches the literature on BO problems with noisy constraints, a relatively new research topic. Recent studies have looked at single objective constrained BO problems for expected improvement (EI) policies (Gelbart et al. 2014, Gramacy 2016, Letham et al., 2019). They applied a GP prior directly on all the solutions and implicitly assumed known noise distributions, which are not applicable to high-dimensional problems with unknown variances. In contrast, our parametric GPR model and the variance learning scheme enable learning under noises for high-dimensional problems. Interestingly, these studies multiplied the original acquisition function (i.e., the unconstrained case) by the probability of feasibility conditional on the posterior, but this leads to very cumbersome computation. However, we do not add this complexity since unlike the EI policy, the KG policy can choose an unsampled decision for final implementation. So, our approach is tractable and can have faster fitting of the proposed parametric GPR model by allowing broader exploration, thus identify Pareto optimal solutions faster. Note that the probabilistic constraint in problem (1) is equivalent to restricting the $(1 - \epsilon)$ -quantile of f_3 to be at least τ . Some related study also used Gaussian process models in optimizing quantiles (Wang et al. 2021). Based on similar Gaussian assumption, we can conveniently predict feasibility using the mean and variance estimates of f_3 . We demonstrate the effectiveness of our approach using both test problems and a real-world urban transportation problem.

The remainder of the paper is organized as follows. We present a literature review of the related SO methods in Section 2. A network-wide signal timing problem under uncertainties (NSTPU) is formulated in Section 3 as an application of model (1). In Section 4, we design parametric GPR and KG sampling policy for NSTP. We apply the proposed method to a test problem and discuss the results in Section 5. Finally, some important findings are drawn in Section 6.